

17-Feb-2026, Request for Changes for V1.2.1 SaleAnto
Deadline: 19-Feb-2026

SaleAnto V1.2.1

Complete User Flow & Developer Instructions

Document Version: 1.2.1 | Date: 16 February 2026

Classification: Internal — Engineering & Product Team

Table of Contents

1. Document Overview & Purpose
2. System Architecture Summary
3. Role Definitions & Permissions Matrix
4. Complete User Flow — 10 Stages
 - 4.1 Stage 1: Strategy & Content Pillar Setup
 - 4.2 Stage 2: Idea Generation & Trending Layer
 - 4.3 Stage 3: Draft Creation & Checklist
 - 4.4 Stage 4: Copy & Caption Generation
 - 4.5 Stage 5: Hashtag Generation Engine
 - 4.6 Stage 6: Creative Assets (Image, Carousel, Video)
 - 4.7 Stage 7: Review & Approval Pipeline
 - 4.8 Stage 8: Scheduling & Publishing
 - 4.9 Stage 9: Post-Publish Analytics & Quick Stats
 - 4.10 Stage 10: Learning Loop & Repurposing
5. Admin & Governance Flow
6. Notification System Specification
7. Generation Limits & Rate Controls
8. Database Schema Requirements
9. API Endpoints Specification
10. Frontend Component Map
11. Phase Delivery Roadmap
12. Acceptance Criteria & QA Checklist

1. Document Overview & Purpose

This document defines the complete user flow and technical implementation instructions for SaleAnto V1.2. It serves as the single source of truth for developers, designers, and QA engineers building the content creation, scheduling, and analytics pipeline.

Scope: This document covers the full journey from strategy setup through post-publish learning loops, including all 19 previously identified missing features now integrated into the flow.

Key Changes from V1.1

- Added Content Pillar/Theme mapping to ideation
- Added complete Hashtag Generation Engine (was entirely missing)
- Added platform-specific copy adaptation for multi-platform posts
- Added auto-resize/reformat for creatives per platform
- Added Best Time Suggestion engine at scheduling
- Added Post-Level Quick Stats (24h/48h snapshots)
- Added Content Repurposing triggers for top performers
- Added Comment/Reply management in daily flow
- Added A/B caption tagging for analytics feedback
- Added Draft Checklist Widget for completion tracking
- Added Alt Text auto-generation for accessibility
- Added Approval SLA Escalation (24h auto-escalate to Owner)
- Added Competitor Inspiration Input in ideation
- Added Trending Topics Layer for real-time relevance
- Added Creative Version History
- Added Cross-Platform Conflict Check at scheduling
- Added Bulk Calendar Scheduling View
- Added Character Count & Live Preview in caption editor
- Added Brand Template Overlay for generated creatives

2. System Architecture Summary

The SaleAnto system follows a modular architecture with the following core services:

Service	Responsibility	Tech Stack
Auth Service	User registration, login, OAuth tokens, role management	JWT + OAuth2
Workspace Service	Workspace CRUD, brand management, team management	REST API
Brand DNA Service	Website crawl, PDF extraction, LLM DNA generation	LLM + Crawler
Ideation Engine	Idea generation, trending topics, competitor analysis	LLM + Trend API
Content Service	Drafts, captions, hashtags, copy adaptation	LLM + NLP
Creative Service	Image gen, video gen, resize, templates, alt text	Image/Video APIs
Approval Service	Workflow states, SLA tracking, escalation	State Machine
Scheduler Service	Queue management, platform publishing, retry logic	Job Queue + Cron
Analytics Service	Metrics sync, weekly reports, learning loop	Platform APIs + DB
Notification Service	In-app, email, Slack/WhatsApp notifications	Event Bus
Messenger Service	Comment monitoring, AI/human reply routing	Webhook + LLM

3. Role Definitions & Permissions Matrix

Role	Description	Key Permissions
Owner	Workspace creator. Full control.	Billing, integrations, permissions, publishing override, all admin rights
Admin	Operations manager.	Brand management, approvals, templates, reports, user role assignment
Creator	Content producer.	Generate ideas, create drafts, generate captions/images/hashtags, submit for approval
Approver	Quality gate.	Approve/reject/request changes on submitted content
Publisher	Granted permission (not a base role).	Schedule and publish approved content
Viewer	Read-only stakeholder.	View analytics, reports, content calendar (no edit rights)

RULE: Publishing requires Owner, Admin, or a user explicitly granted 'Publisher' permission. Creator role alone cannot publish.

4. Complete User Flow — 10 Stages

The following 10 stages define the complete content lifecycle from strategy to learning. Each stage includes user actions, system behavior, developer instructions, database requirements, API endpoints, and acceptance criteria.

4.1 Stage 1: Strategy & Content Pillar Setup

Purpose: Before generating any content, users define their content strategy by setting up content pillars and themes. This ensures the content calendar is balanced and strategic, not random.

User Actions

- Navigate to Strategy Hub (accessible from sidebar)
- Create Content Pillars (e.g., Educational, Promotional, Engagement, Storytelling, Behind-the-scenes)
- Set pillar distribution ratio (e.g., 40% Educational, 25% Promotional, 20% Engagement, 15% Storytelling)
- Optionally add Competitor Inspiration URLs (up to 5 competitor handles/URLs per brand)
- Save strategy

System Behavior

- Store content pillars linked to brand_id
- If competitor URLs provided: crawl public posts (last 30 days), extract top hooks/angles/formats, store as competitor_insights
- Auto-tag future ideas with matching pillars based on content analysis
- Dashboard shows pillar distribution compliance (actual vs target)

Developer Instructions

```
// Database Tables Required:
content_pillars {
  id: UUID (PK)
  brand_id: UUID (FK → brands)
  name: VARCHAR(100)
  description: TEXT
  target_percentage: INT (0-100)
  color_code: VARCHAR(7) // for UI calendar display
  is_active: BOOLEAN DEFAULT true
  created_at: TIMESTAMP
  updated_at: TIMESTAMP
}

competitor_profiles {
  id: UUID (PK)
  brand_id: UUID (FK → brands)
  platform: ENUM('twitter','linkedin','facebook','instagram')
  handle_or_url: VARCHAR(500)
  last_crawled_at: TIMESTAMP
  created_at: TIMESTAMP
}

competitor_insights {
  id: UUID (PK)
  competitor_profile_id: UUID (FK → competitor_profiles)
  hook_text: TEXT
  angle: VARCHAR(200)
  format_type: VARCHAR(50)
  engagement_score: FLOAT // estimated
  extracted_at: TIMESTAMP
}
```

API Endpoints

Method	Endpoint	Description	Auth
POST	/api/v1/brands/{id}/pillars	Create content pillar	Owner, Admin
GET	/api/v1/brands/{id}/pillars	List all pillars for brand	All roles
PUT	/api/v1/brands/{id}/pillars/{pillar_id}	Update pillar	Owner, Admin
DELETE	/api/v1/brands/{id}/pillars/{pillar_id}	Soft-delete pillar	Owner, Admin
POST	/api/v1/brands/{id}/competitors	Add competitor profile	Owner, Admin, Creator
POST	/api/v1/brands/{id}/competitors/crawl	Trigger competitor crawl	Owner, Admin
GET	/api/v1/brands/{id}/competitors/insights	Get competitor insights	All roles
GET	/api/v1/brands/{id}/pillar-compliance	Get actual vs target distribution	Owner, Admin

Acceptance Criteria

- User can create 2–10 content pillars per brand
- Pillar percentages must sum to 100%
- Competitor crawl completes within 60 seconds
- Pillar compliance chart renders on Strategy Hub dashboard
- Auto-tagging of ideas to pillars works with >80% accuracy

4.2 Stage 2: Idea Generation & Trending Layer

Purpose: Generate content ideas informed by Brand DNA, content pillars, competitor insights, and real-time trending topics. This is the daily 5–10 minute workflow for creators.

User Actions

- Open Ideas Hub
- Select: Platform → Goal → Format → Language → Content Pillar (optional filter)

- Click "Generate Ideas"
- View ideas in two panels: AI-Generated Ideas (left) + Trending Topics (right)
- Actions per idea: Save / Skip / Regenerate / Add to Calendar
- Saved ideas auto-tagged with pillar, persona, hook type

System Behavior

- Default: 20 ideas per run (options: 10/20/50)
- LLM generates ideas using: Brand DNA + selected pillar + goal + persona + format constraints
- Trending Layer: Pull trending topics from platform APIs (Twitter Trends, LinkedIn trending, Google Trends)
- Match trending topics against brand relevance score (0-1). Show only score > 0.4
- Each idea includes: headline, hook, angle, suggested format, target persona, pillar tag, estimated engagement tier (low/mid/high)

Developer Instructions

// Database Tables:

```
ideas {
  id: UUID (PK)
  brand_id: UUID (FK → brands)
  workspace_id: UUID (FK → workspaces)
  created_by: UUID (FK → users)
  headline: VARCHAR(300)
  hook: TEXT
  angle: VARCHAR(200)
  format_type: ENUM('static', 'carousel', 'reel', 'thread', 'story')
  platform: ENUM('twitter', 'linkedin', 'facebook', 'instagram')
  goal: ENUM('leads', 'growth', 'authority')
  pillar_id: UUID (FK → content_pillars, NULLABLE)
  persona_id: UUID (FK → personas, NULLABLE)
  engagement_tier: ENUM('low', 'mid', 'high')
  status: ENUM('generated', 'saved', 'skipped', 'converted') DEFAULT 'generated'
  source: ENUM('ai_generated', 'trending', 'competitor_inspired') DEFAULT 'ai_generated'
  trending_topic_ref: VARCHAR(500) // if sourced from trending
  metadata_json: JSONB // hook type, offer, extra context
  created_at: TIMESTAMP
  updated_at: TIMESTAMP
}
```

```
trending_cache {
  id: UUID (PK)
  platform: ENUM('twitter', 'linkedin', 'google')
  topic: VARCHAR(500)
  volume_score: FLOAT
  region: VARCHAR(100)
  fetched_at: TIMESTAMP
  expires_at: TIMESTAMP
}
```

```
// LLM Prompt Template (store in prompt_templates table):
// System: You are a social media strategist for {brand_name}.
// Context: Brand DNA: {brand_dna_json}, Pillar: {pillar_name},
```

```
// Goal: {goal}, Persona: {persona_description},
// Platform: {platform}, Format: {format}
// Task: Generate {count} content ideas. For each: headline, hook,
// angle, engagement_tier estimate.
// Rules: Follow Do/Don't rules: {do_dont_rules}.
// Avoid banned claims: {banned_claims}.
```

API Endpoints

Method	Endpoint	Description	Auth
POST	/api/v1/ideas/generate	Generate ideas (LLM call)	Creator, Admin, Owner
GET	/api/v1/ideas?status=saved&brand_id=X	List saved ideas	All roles
PATCH	/api/v1/ideas/{id}/status	Update idea status (save/skip)	Creator, Admin, Owner
POST	/api/v1/ideas/{id}/add-to-calendar	Add idea to content calendar	Creator, Admin, Owner
GET	/api/v1/trending?platform=X®ion=Y	Get trending topics	All roles
POST	/api/v1/ideas/{id}/regenerate	Regenerate single idea	Creator, Admin, Owner

Acceptance Criteria

- 20 ideas generated in under 8 seconds
- Trending panel loads within 3 seconds
- Each idea correctly tagged to content pillar
- Save/skip actions persist immediately
- Calendar view shows saved ideas on correct dates
- Generation respects daily workspace limits (default 200/day)

4.3 Stage 3: Draft Creation & Checklist

Purpose: Convert a saved idea into a workable draft with a visible completion checklist so the creator knows exactly what's needed before submission.

User Actions

- Click "Create Draft" on any saved idea
- Post Editor opens with pre-filled metadata (hook, angle, persona, format, goal, pillar)
- Draft Checklist Widget visible in sidebar showing completion status

Draft Checklist Widget — Required Items

Checklist Item	Status Logic	Required?
Caption / Copy	At least 1 caption variant generated and selected	YES
Hashtags	Hashtags generated and confirmed (or manually added)	YES (Instagram/LinkedIn), Optional (Twitter)
Creative Asset	At least 1 image/video attached or generated	YES
Alt Text	Alt text exists for all images	YES
Platform Mapping	Each target platform has assigned copy + creative	YES (if multi-platform)
CTA	Call-to-action defined	Recommended
Schedule Slot	Date/time selected	Optional at draft stage

Developer Instructions

```
// Database Tables:
drafts {
  id: UUID (PK)
  idea_id: UUID (FK → ideas)
  brand_id: UUID (FK → brands)
  workspace_id: UUID (FK → workspaces)
  created_by: UUID (FK → users)
  status: ENUM('draft','pending_approval','approved','changes_requested',
    'rejected','scheduled','published','failed') DEFAULT 'draft'
  pillar_id: UUID (FK → content_pillars, NULLABLE)
  persona_id: UUID (FK → personas, NULLABLE)
  goal: ENUM('leads','growth','authority')
```

```

hook: TEXT
angle: VARCHAR(200)
format_type: ENUM('static','carousel','reel','thread','story')
cta_text: VARCHAR(300)
metadata_json: JSONB
checklist_status: JSONB // {"caption": true, "hashtags": false, ...}
created_at: TIMESTAMP
updated_at: TIMESTAMP
submitted_at: TIMESTAMP
approved_at: TIMESTAMP
scheduled_at: TIMESTAMP
published_at: TIMESTAMP
}

// Frontend Component: DraftChecklistWidget
// - Renders as a sticky sidebar in the Post Editor
// - Reads checklist_status JSONB from draft record
// - Updates in real-time as user completes each item
// - Shows green check / red X / yellow warning per item
// - "Submit for Approval" button DISABLED until all required items = true

```

API Endpoints

Method	Endpoint	Description	Auth
POST	/api/v1/drafts	Create draft from idea	Creator, Admin, Owner
GET	/api/v1/drafts/{id}	Get draft with full details	Creator, Admin, Owner, Approver
PATCH	/api/v1/drafts/{id}	Update draft fields	Creator (own), Admin, Owner
GET	/api/v1/drafts/{id}/checklist	Get checklist status	Creator, Admin, Owner
PATCH	/api/v1/drafts/{id}/checklist	Update checklist item	System (auto-triggered)

Acceptance Criteria

- Draft auto-populates all metadata from source idea
- Checklist widget renders in real-time as items complete
- Submit button disabled until all required checklist items are green
- Creator cannot bypass checklist (server-side validation on submit)

4.4 Stage 4: Copy & Caption Generation

Purpose: Generate platform-optimized caption variants with A/B tagging support, real-time character count, and live platform preview.

User Actions

- Inside Post Editor, click 'Generate Captions'
- Select tone override (or use Brand DNA default)
- System generates 3 caption variants (configurable up to 5)
- If multi-platform post: system generates platform-specific adaptations automatically
- User picks primary caption, optionally tags 2 variants as A/B test
- Add/edit CTA per variant
- Character count shows live per platform with color coding (green = safe, yellow = near limit, red = over)
- Live Preview panel shows how post will appear on each target platform

Platform Character Limits (Enforced)

Platform	Character Limit	Optimal Length	Notes
Twitter/X	280	200-250	Include space for hashtags
LinkedIn	3,000	800-1,300	First 210 chars visible before 'see more'
Facebook	63,206	40-80	Shorter posts get higher engagement
Instagram	2,200	138-150	First 125 chars visible in feed

Platform-Specific Adaptation Logic

- Twitter: Compress to ≤280 chars, conversational tone, 1 CTA max, 2-3 hashtags inline
- LinkedIn: Professional tone, longer narrative, hook in first 210 chars, CTA at end
- Instagram: Visual-first language, emojis allowed, hashtags in caption body or first comment option
- Facebook: Casual, question-led hooks, short and punchy

Developer Instructions

```
// Database Tables:
captions {
```

```

id: UUID (PK)
draft_id: UUID (FK → drafts)
platform: ENUM('twitter','linkedin','facebook','instagram','all')
variant_number: INT (1-5)
body: TEXT
cta_text: VARCHAR(300)
tone: VARCHAR(100)
char_count: INT
is_selected: BOOLEAN DEFAULT false
is_ab_test: BOOLEAN DEFAULT false
ab_label: ENUM('A','B', NULL) DEFAULT NULL
generation_prompt_hash: VARCHAR(64) // for debugging
created_at: TIMESTAMP
updated_at: TIMESTAMP
}

// LLM Prompt for Platform Adaptation:
// Input: Original caption + target platform + Brand DNA
// Output: Platform-optimized version respecting char limits, tone, format
// Key rule: First line must be the hook. CTA in last line.

// Frontend Components:
// 1. CaptionEditor - Rich text editor with char counter
// 2. PlatformPreviewPanel - Shows mock-up of post per platform
// 3. ABToggle - Click to tag variant as A or B
// 4. ToneSelector - Dropdown with Brand DNA tones + custom override

```

API Endpoints

Method	Endpoint	Description	Auth
POST	/api/v1/drafts/{id}/captions/generate	Generate caption variants	Creator, Admin, Owner
POST	/api/v1/drafts/{id}/captions/adapt	Adapt caption to specific platform	Creator, Admin, Owner
PATCH	/api/v1/captions/{id}	Edit caption text	Creator (own), Admin, Owner
PATCH	/api/v1/captions/{id}/select	Select/deselect as primary	Creator (own), Admin, Owner
PATCH	/api/v1/captions/{id}/ab-tag	Tag as A/B variant	Creator (own), Admin, Owner
GET	/api/v1/drafts/{id}/captions	List all captions for draft	All roles

GET	/api/v1/captions/{id}/preview/{platform}	Get platform preview data	All roles
-----	--	---------------------------	-----------

Acceptance Criteria

- 3 variants generated in under 5 seconds
- Platform adaptation produces correctly sized output per platform
- Character counter updates in real-time with correct color coding
- Live preview accurately reflects platform display
- A/B tagging persists and links to analytics for post-publish comparison
- Compliance rules (banned keywords, do/don't) enforced — violations flagged before save

4.5 Stage 5: Hashtag Generation Engine

NEW: This stage was entirely missing from V1.1 and is a critical addition.

Purpose: Auto-generate optimized hashtags organized by reach tier, with platform-specific quantities and placement options.

User Actions

- After caption is confirmed, click 'Generate Hashtags'
- System generates hashtags in 3 tiers: High-Volume (reach), Mid-Volume (relevance), Niche (authority)
- User can: add/remove individual hashtags, save as reusable Hashtag Group, select placement (in-caption vs first-comment for Instagram)
- Banned hashtag filter auto-removes off-brand or risky tags

Default Hashtag Counts by Platform

Platform	Default Count	Max Count	Placement
Instagram	20	30	Caption body OR first comment (user choice)
LinkedIn	5	10	End of caption
Twitter/X	3	5	Inline within caption

Facebook	3	5	End of caption
----------	---	---	----------------

Hashtag Tier Logic

- **Reach** — High-Volume (30%):
 - Tags with >500K posts. Purpose: maximize impressions.
- **Relevance** — Mid-Volume (40%):
 - Tags with 50K–500K posts. Purpose: target the right audience.
- **Authority** — Niche (30%):
 - Tags with <50K posts. Purpose: dominate in specific category.

Developer Instructions

```
// Database Tables:
hashtags {
  id: UUID (PK)
  draft_id: UUID (FK → drafts)
  platform: ENUM('twitter','linkedin','facebook','instagram')
  tag: VARCHAR(200) // without #
  tier: ENUM('high_volume','mid_volume','niche')
  estimated_volume: INT
  is_selected: BOOLEAN DEFAULT true
  placement: ENUM('inline','end_of_caption','first_comment') DEFAULT 'end_of_caption'
  created_at: TIMESTAMP
}

hashtag_groups {
  id: UUID (PK)
  brand_id: UUID (FK → brands)
  name: VARCHAR(100)
  tags: TEXT[] // array of hashtag strings
  created_by: UUID (FK → users)
  created_at: TIMESTAMP
}

banned_hashtags {
  id: UUID (PK)
  brand_id: UUID (FK → brands)
  tag: VARCHAR(200)
  reason: VARCHAR(300)
  added_by: UUID (FK → users)
  created_at: TIMESTAMP
}

// LLM Prompt for Hashtag Generation:
// Input: Caption text + Brand DNA + platform + pillar + persona
// Output: JSON array of {tag, tier, estimated_volume}
// Rules: Exclude banned list. Mix 30/40/30 tier ratio.
// All tags must be relevant to content, not generic spam.
```

API Endpoints

Method	Endpoint	Description	Auth
POST	/api/v1/drafts/{id}/hashtags/generate	Generate hashtags for draft	Creator, Admin, Owner
GET	/api/v1/drafts/{id}/hashtags	List hashtags for draft	All roles
PATCH	/api/v1/hashtags/{id}	Toggle select/deselect, change placement	Creator, Admin, Owner
POST	/api/v1/brands/{id}/hashtag-groups	Save reusable hashtag group	Creator, Admin, Owner
GET	/api/v1/brands/{id}/hashtag-groups	List saved hashtag groups	All roles
POST	/api/v1/brands/{id}/banned-hashtags	Add banned hashtag	Admin, Owner

Acceptance Criteria

- Hashtags generated in under 3 seconds
- Tier distribution matches 30/40/30 ratio ($\pm 5\%$)
- Banned hashtags never appear in results
- Instagram first-comment placement works on publish
- Saved hashtag groups persist and are reusable across drafts

4.6 Stage 6: Creative Assets (Image, Carousel, Video)

Purpose: Generate or upload visual assets with auto-resize per platform, brand template overlays, alt text generation, and version history.

User Actions

- In Post Editor, click 'Generate Creative' or 'Upload'
- For generation: system uses caption + Brand DNA + style presets to create images
- For carousel: system auto-splits content into slides
- For video/reel: upload only (Phase 1), generation in Phase 2
- User selects brand template overlay (optional)

- System auto-generates alt text for each image
- System auto-resizes to all target platform dimensions
- User can regenerate (previous versions kept in history)

Auto-Resize Dimensions

Platform	Format	Dimensions	Aspect Ratio
Instagram	Feed Post	1080 × 1080	1:1
Instagram	Story/Reel	1080 × 1920	9:16
Instagram	Carousel	1080 × 1080	1:1
LinkedIn	Feed Post	1200 × 627	1.91:1
Twitter/X	Feed Post	1200 × 675	16:9
Facebook	Feed Post	1200 × 630	1.91:1
Facebook	Story	1080 × 1920	9:16

Developer Instructions

```
// Database Tables:
creative_assets {
  id: UUID (PK)
  draft_id: UUID (FK → drafts)
  asset_type: ENUM('image','carousel_slide','video','voiceover')
  file_url: VARCHAR(1000) // S3/CDN URL
  thumbnail_url: VARCHAR(1000)
  original_dimensions: VARCHAR(20) // "1080x1080"
  file_size_bytes: BIGINT
  alt_text: TEXT
  brand_template_id: UUID (FK → brand_templates, NULLABLE)
  version: INT DEFAULT 1
  is_current: BOOLEAN DEFAULT true
  metadata_json: JSONB // generation params, model used, etc.
  created_at: TIMESTAMP
}

asset_platform_variants {
  id: UUID (PK)
  asset_id: UUID (FK → creative_assets)
  platform: ENUM('twitter','linkedin','facebook','instagram')
  format_label: VARCHAR(50) // 'feed','story','reel'
```

```

file_url: VARCHAR(1000)
dimensions: VARCHAR(20)
created_at: TIMESTAMP
}

brand_templates {
id: UUID (PK)
brand_id: UUID (FK → brands)
name: VARCHAR(100)
template_file_url: VARCHAR(1000) // PSD/Figma/SVG template
logo_position: ENUM('top_left','top_right','bottom_left','bottom_right','center')
font_family: VARCHAR(100)
primary_color: VARCHAR(7)
secondary_color: VARCHAR(7)
is_active: BOOLEAN DEFAULT true
created_at: TIMESTAMP
}

creative_version_history {
id: UUID (PK)
asset_id: UUID (FK → creative_assets)
version: INT
file_url: VARCHAR(1000)
generation_params: JSONB
created_at: TIMESTAMP
}

// Auto-Resize Pipeline (background job):
// 1. On asset upload/generation → trigger resize job
// 2. For each target platform in draft.platforms → generate variant
// 3. Store variants in asset_platform_variants
// 4. Use sharp (Node) or Pillow (Python) for resize + crop

// Alt Text Generation:
// POST image to LLM vision endpoint
// Prompt: "Generate descriptive alt text for accessibility. Max 125 chars.
// Context: {brand_name}, {caption_summary}"
// Store in creative_assets.alt_text

```

API Endpoints

Method	Endpoint	Description	Auth
POST	/api/v1/drafts/{id}/assets/generate	Generate image via AI	Creator, Admin, Owner
POST	/api/v1/drafts/{id}/assets/upload	Upload asset file	Creator, Admin, Owner
POST	/api/v1/assets/{id}/resize	Trigger platform resize	System (auto)

POST	/api/v1/assets/{id}/alt-text	Generate alt text	System (auto) or manual
POST	/api/v1/assets/{id}/apply-template	Apply brand template overlay	Creator, Admin, Owner
GET	/api/v1/assets/{id}/versions	Get version history	Creator, Admin, Owner
POST	/api/v1/assets/{id}/regenerate	Regenerate (saves old version)	Creator, Admin, Owner
GET	/api/v1/drafts/{id}/assets	List all assets for draft	All roles

Acceptance Criteria

- Image generation completes in under 15 seconds
- Auto-resize produces correct dimensions for all target platforms
- Alt text generated for every image (≤125 chars)
- Brand template overlay correctly positions logo, colors, fonts
- Version history preserves all previous generations
- Carousel generation correctly splits content into 2-10 slides

4.7 Stage 7: Review & Approval Pipeline

Purpose: Mandatory quality gate. All content must pass approval before scheduling (default ON, configurable). Includes SLA escalation.

User Actions

- Creator clicks "Submit for Approval" (only enabled when checklist is complete)
- Approver receives notification
- Approver sees: post preview, all caption variants, creative assets, hashtags, compliance checklist, metadata summary
- Approver actions: Approve / Request Changes (must comment) / Reject (must select reason)
- If changes requested: Creator notified, edits, resubmits

SLA Escalation Rules

Time Pending	Action	Notification Target
12 hours	Reminder notification	Assigned Approver
24 hours	Escalation notification	Owner + Admin
48 hours	Auto-flag as 'Stale' in dashboard	Owner (decision required)

Developer Instructions

```
// Database Tables:
approval_logs {
  id: UUID (PK)
  draft_id: UUID (FK → drafts)
  action: ENUM('submitted','approved','changes_requested','rejected','escalated')
  acted_by: UUID (FK → users)
  comment: TEXT
  rejection_reason: ENUM('off_brand','compliance_issue','quality','factual_error',
                        'timing','other', NULL)
  created_at: TIMESTAMP
}
```

```
// SLA Cron Job (runs every hour):
// 1. Query drafts WHERE status = 'pending_approval'
// AND submitted_at < NOW() - INTERVAL '12 hours'
// AND no reminder sent
// 2. Send reminder to approver
// 3. At 24h → notify Owner + Admin
// 4. At 48h → flag as 'stale', require Owner decision
```

```
// Approval State Machine:
// draft → pending_approval → approved → scheduled → published
//           → changes_requested → draft (loop back)
//           → rejected (terminal, can clone to new draft)
```

API Endpoints

Method	Endpoint	Description	Auth
POST	/api/v1/drafts/{id}/submit	Submit for approval	Creator (own)
POST	/api/v1/drafts/{id}/approve	Approve draft	Approver, Admin, Owner

POST	/api/v1/drafts/{id}/request-changes	Request changes (with comment)	Approver, Admin, Owner
POST	/api/v1/drafts/{id}/reject	Reject (with reason)	Approver, Admin, Owner
GET	/api/v1/approvals/pending	List all pending approvals	Approver, Admin, Owner
GET	/api/v1/drafts/{id}/approval-log	Get approval history	All roles

Acceptance Criteria

- Submit blocked if checklist incomplete (server-side validation)
- Approver notification delivered within 30 seconds of submission
- 12h/24h/48h SLA escalation triggers correctly
- Approval log records all state transitions with timestamps
- Rejected drafts can be cloned to new drafts

4.8 Stage 8: Scheduling & Publishing

Purpose: Schedule approved content with best-time suggestions, bulk calendar view, cross-platform conflict detection, and reliable publishing.

User Actions

- After approval, Creator/Publisher opens Scheduling view
- System suggests optimal posting times per platform (based on past analytics or industry defaults)
- User selects date/time per platform (or accepts suggestion)
- User confirms platform-to-asset mapping (each platform gets correct copy + creative + hashtags)
- Bulk Calendar View: drag-and-drop for weekly/monthly scheduling
- Cross-platform conflict check warns if same post goes to multiple platforms at identical time

Best Time Suggestion Logic

- If workspace has 30+ published posts with analytics: use own data (top engagement time slots)
- If <30 posts: use industry-standard defaults per platform

- Suggestion shows as highlighted slots on calendar + tooltip with reasoning

Developer Instructions

```
// Database Tables:
scheduled_posts {
  id: UUID (PK)
  draft_id: UUID (FK → drafts)
  platform: ENUM('twitter','linkedin','facebook','instagram')
  caption_id: UUID (FK → captions)
  asset_variant_id: UUID (FK → asset_platform_variants)
  hashtag_placement: ENUM('inline','end_of_caption','first_comment')
  scheduled_at: TIMESTAMP
  timezone: VARCHAR(50)
  status: ENUM('scheduled','publishing','published','failed','cancelled')
  publish_result_json: JSONB // platform post ID, URL, errors
  retry_count: INT DEFAULT 0
  max_retries: INT DEFAULT 3
  created_by: UUID (FK → users)
  created_at: TIMESTAMP
  published_at: TIMESTAMP
}

best_time_suggestions {
  id: UUID (PK)
  brand_id: UUID (FK → brands)
  platform: ENUM('twitter','linkedin','facebook','instagram')
  day_of_week: INT (0=Mon, 6=Sun)
  hour_utc: INT (0-23)
  score: FLOAT // engagement score
  source: ENUM('own_data','industry_default')
  computed_at: TIMESTAMP
}

// Publishing Pipeline (Job Queue):
// 1. Cron checks scheduled_posts WHERE scheduled_at <= NOW()
// AND status = 'scheduled'
// 2. For each: call platform API to publish
// 3. If success: status = 'published', store platform post ID
// 4. If fail: increment retry_count, retry after 5min (max 3)
// 5. If max retries exceeded: status = 'failed', notify Creator + Admin

// Cross-Platform Conflict Detection:
// On schedule creation: query scheduled_posts WHERE draft_id = X
// If any two have scheduled_at within 15 minutes → warn user
// Suggest staggering by 30-60 minutes

// Bulk Calendar Frontend:
// FullCalendar.js or similar library
// Drag-and-drop support for rescheduling
// Color-coded by platform
// Visual indicators for: suggested times (highlighted), conflicts (red border)
```

API Endpoints

Method	Endpoint	Description	Auth
POST	/api/v1/drafts/{id}/schedule	Schedule post for platforms	Creator, Publisher, Admin, Owner
GET	/api/v1/schedule/calendar?range=week	Get calendar view data	All roles
PATCH	/api/v1/scheduled-posts/{id}	Reschedule (change time)	Creator, Publisher, Admin, Owner
DELETE	/api/v1/scheduled-posts/{id}	Cancel scheduled post	Creator (own), Admin, Owner
GET	/api/v1/brands/{id}/best-times	Get best time suggestions	All roles
POST	/api/v1/schedule/conflict-check	Check for scheduling conflicts	System (auto on schedule)

Acceptance Criteria

- Publishing executes within 60 seconds of scheduled time
- Failed posts retry 3 times with 5-min intervals
- Conflict detection triggers for posts within 15 minutes of each other
- Best time suggestions display correctly on calendar
- Bulk drag-and-drop rescheduling works without page reload
- Instagram first-comment hashtags post correctly after main post

4.9 Stage 9: Post-Publish Analytics & Quick Stats

Purpose: Provide immediate post-performance feedback (24h/48h snapshots) plus weekly comprehensive reports, and enable comment/reply management.

Post-Level Quick Stats (24h / 48h)

- After publishing, the post card shows a quick stats badge:
 - Impressions, Reach, Engagement Rate, Clicks, Comments, Shares
- First snapshot at 24h, second at 48h, then daily sync

- Color-coded performance indicator: Green (above average), Yellow (average), Red (below)

Comment & Reply Management

- System monitors published posts for new comments via platform webhooks/polling
- Comments appear in Messenger module inbox
- AI/Human reply toggle (set during onboarding, configurable per brand)
- AI auto-reply: uses Brand DNA tone + reply style templates
- Human reply: notification sent to Creator, reply from within SaleAnto
- Reply SLA: configurable (default 2 hours for first response)

Weekly Report (Auto-generated)

- Winners and losers (top 3 / bottom 3)
- Best hooks and angles that performed
- Best posting times and formats
- Content pillar performance breakdown
- A/B test results (if any captions were A/B tagged)
- Recommendations for next week
- Next week test plan (10 experiments default)

Developer Instructions

```
// Database Tables:
post_analytics {
  id: UUID (PK)
  scheduled_post_id: UUID (FK → scheduled_posts)
  platform: ENUM('twitter','linkedin','facebook','instagram')
  platform_post_id: VARCHAR(200) // external ID
  snapshot_type: ENUM('24h','48h','daily','weekly')
  impressions: INT
  reach: INT
  engagement_rate: FLOAT
  likes: INT
  comments_count: INT
  shares: INT
  clicks: INT
  saves: INT
  profile_visits: INT
  data_json: JSONB // platform-specific extra metrics
  fetched_at: TIMESTAMP
}

post_comments {
  id: UUID (PK)
  scheduled_post_id: UUID (FK → scheduled_posts)
  platform: ENUM('twitter','linkedin','facebook','instagram')
  platform_comment_id: VARCHAR(200)
  author_name: VARCHAR(200)
  author_handle: VARCHAR(200)
  body: TEXT
  sentiment: ENUM('positive','neutral','negative')
```

```

replied: BOOLEAN DEFAULT false
reply_type: ENUM('ai','human', NULL)
reply_body: TEXT
replied_at: TIMESTAMP
fetched_at: TIMESTAMP
}

```

```

weekly_reports {
id: UUID (PK)
brand_id: UUID (FK → brands)
workspace_id: UUID (FK → workspaces)
week_start: DATE
week_end: DATE
winners: JSONB // [{post_id, metric, value}]
losers: JSONB
best_hooks: JSONB
best_times: JSONB
pillar_performance: JSONB
ab_test_results: JSONB
recommendations: TEXT[]
test_plan: JSONB
generated_at: TIMESTAMP
}

```

```

// Analytics Sync Cron (runs every 6 hours):
// 1. Query all published posts from last 7 days
// 2. For each: call platform API for metrics
// 3. Store snapshot in post_analytics
// 4. At 24h mark: store '24h' snapshot
// 5. At 48h mark: store '48h' snapshot

// Weekly Report Generator (runs Sunday midnight):
// 1. Aggregate post_analytics for the week
// 2. Rank by engagement_rate for winners/losers
// 3. Extract hook patterns from winning captions
// 4. Calculate pillar distribution vs targets
// 5. Compare A/B variants
// 6. LLM generates recommendations + test plan
// 7. Store in weekly_reports
// 8. Notify Owner + Admin

```

API Endpoints

Method	Endpoint	Description	Auth
GET	/api/v1/posts/{id}/stats	Get post quick stats	All roles
GET	/api/v1/posts/{id}/comments	Get comments for post	Creator, Admin, Owner

POST	/api/v1/comments/{id}/reply	Reply to comment	Creator, Admin, Owner
POST	/api/v1/comments/{id}/ai-reply	Trigger AI reply	Creator, Admin, Owner
GET	/api/v1/brands/{id}/weekly-report?week=X	Get weekly report	Admin, Owner
GET	/api/v1/brands/{id}/analytics/dashboard	Get analytics dashboard data	All roles
GET	/api/v1/brands/{id}/ab-results	Get A/B test results	All roles

Acceptance Criteria

- 24h quick stats badge appears on post card within 25 hours of publishing
- Comments sync within 15 minutes of being posted on platform
- AI reply generates within 5 seconds and matches Brand DNA tone
- Weekly report auto-generates every Sunday by 2:00 AM
- A/B results correctly compare tagged variants
- Pillar performance shows actual vs target distribution

4.10 Stage 10: Learning Loop & Content Repurposing

Purpose: Close the feedback loop by feeding performance data back into the system, and actively prompt repurposing of top-performing content.

Learning Loop (Automated)

- Weekly report insights auto-update Best Time Suggestions
- Winning hooks and angles boost priority in next idea generation cycle
- Poorly performing pillars get flagged for strategy review
- A/B results update default tone/format preferences
- Phase 4 (future): fully automated next-week plan generation

Content Repurposing (Active Prompts)

- When a post is flagged as a "winner" (top 20% engagement), system triggers repurpose prompt:

- "This post performed well. Repurpose as: Carousel / Thread / Reel / Email / Blog outline?"
- User selects target format → system creates new draft pre-filled with adapted content
- Repurposed draft links back to original for tracking

Developer Instructions

```
// Database Tables:
learning_signals {
  id: UUID (PK)
  brand_id: UUID (FK → brands)
  signal_type: ENUM('best_hook','best_time','best_format','best_pillar',
    'worst_hook','worst_time','worst_format','ab_winner')
  reference_id: UUID // post or caption ID
  data_json: JSONB
  applied: BOOLEAN DEFAULT false // has this signal been used in generation?
  created_at: TIMESTAMP
}

repurposed_content {
  id: UUID (PK)
  original_draft_id: UUID (FK → drafts)
  new_draft_id: UUID (FK → drafts)
  repurpose_format: ENUM('carousel','thread','reel','email','blog_outline')
  created_at: TIMESTAMP
}

// Learning Signal Pipeline (weekly):
// 1. After weekly_report generated → extract signals
// 2. Store in learning_signals
// 3. Next idea generation includes learning_signals as context
// 4. LLM prompt appends: "Prioritize hooks similar to: {winning_hooks}"

// Repurposing Trigger:
// 1. After 48h snapshot → check if engagement_rate > brand_average * 1.5
// 2. If yes → flag as 'winner'
// 3. Create in-app notification: "Repurpose this winner?"
// 4. On user action → create new draft with adapted content via LLM
```

API Endpoints

Method	Endpoint	Description	Auth
GET	/api/v1/brands/{id}/learning-signals	Get active learning signals	Admin, Owner
POST	/api/v1/posts/{id}/repurpose	Create repurposed draft	Creator, Admin, Owner
GET	/api/v1/brands/{id}/winners	Get top performing posts	All roles

Acceptance Criteria

- Learning signals extracted within 1 hour of weekly report generation
- Repurpose prompt appears within 50 hours of publishing for qualifying posts
- Repurposed drafts correctly adapt content to new format
- Link between original and repurposed content is trackable in analytics

5. Admin & Governance Flow

5.1 Admin Dashboard

- Content pipeline status: Draft → Pending → Approved → Scheduled → Published (with counts)
- Queue health: image/video generation jobs in progress/failed
- Platform token health: green/yellow/red per connected platform
- API usage and cost estimation (LLM calls, image gen calls)
- Pending approval count with SLA status
- Failed jobs list with retry buttons
- Content pillar compliance chart (actual vs target distribution)

5.2 Brand Management

- Edit Brand DNA (re-crawl website, re-upload PDFs)
- Add new brand assets (logos, images, templates)
- Create additional brands (post-onboarding)
- Set per-brand compliance rules and banned keywords
- Manage banned hashtags list
- Configure content pillars and distribution targets

5.3 Template Management

- Caption templates (reusable copy structures)
- Ad templates (Phase 3)
- UGC templates (Phase 3)
- Email templates (Phase 3)
- Brand creative templates (logo overlay, color schemes)
- Style presets (tone + format combinations)

5.4 Permissions Management

- Assign roles per user per workspace
- Grant Publisher permission to specific users
- Configure who can: approve, publish, export, manage brands
- Audit log of all permission changes

5.5 System Health

- Retry failed jobs (individual or bulk)
- Pause/resume bulk generation
- Enable fallback AI provider rules
- Monitor rate limits and queue depth
- Platform API error logs

6. Notification System Specification

Event	Channel	Recipients	Timing
Post submitted for approval	In-app + Email	Assigned Approver	Immediate
Post approved	In-app	Creator	Immediate
Changes requested	In-app + Email	Creator	Immediate
Post rejected	In-app + Email	Creator	Immediate
Approval pending >12h	In-app + Email	Approver	12h after submit
Approval pending >24h (escalation)	In-app + Email	Owner + Admin	24h after submit
Post scheduled	In-app	Creator	Immediate
Post published	In-app	Creator	Immediate

Publish failed	In-app + Email	Creator + Admin	Immediate
Captions ready	In-app	Creator	Immediate
Images ready	In-app	Creator	Immediate
Video rendering started	In-app	Creator	Immediate
Video ready	In-app	Creator	On completion
Batch complete	In-app	Creator	1 summary per batch
Weekly report ready	In-app + Email	Owner + Admin	Sunday
Winner detected	In-app	Creator + Admin	48h post-publish
Repurpose suggestion	In-app	Creator	48h post-publish
New comment on post	In-app	Creator	Within 15 min
Platform token expiring	In-app + Email	Owner + Admin	7 days before
Daily limit approaching (80%)	In-app	Admin	On threshold

7. Generation Limits & Rate Controls

Resource	Default	Max (Configurable)	Notes
Ideas per generation run	20	50	Options: 10/20/50

Caption variants per draft	3	5	—
Images per draft	2	4	—
Videos per draft	1	2	Queue priority based
Ad variations	3	10 (Admin only)	Bulk = admin permission
Email sequences	3 emails	5 emails	Phase 3
Hashtags per platform	See platform table	30 (Instagram max)	—
Generations per day (workspace)	200	Configurable by Owner	Hard safety limit
Video renders per day	20	50	Rate limited to prevent queue overload
Competitor crawls per day	5	10	Avoid rate limiting on platforms

8. Database Schema Requirements — Complete Table List

Below is the complete list of database tables required for V1.2. Detailed column definitions are provided in the relevant stage sections above.

Table Name	Purpose	Defined In
workspaces	Workspace management	Stage: Onboarding

users	User accounts and profiles	Stage: Onboarding
user_roles	Role assignments per workspace	Stage: Onboarding
brands	Brand profiles	Stage: Onboarding
brand_assets	Logos, PDFs, images per brand	Stage: Onboarding
brand_dna	Generated Brand DNA JSON	Stage: Onboarding
platform_connections	OAuth tokens per platform	Stage: Onboarding
content_pillars	Content strategy pillars	Stage 1: Strategy
competitor_profiles	Competitor URLs/handles	Stage 1: Strategy
competitor_insights	Extracted competitor data	Stage 1: Strategy
ideas	Generated and saved ideas	Stage 2: Ideation
trending_cache	Cached trending topics	Stage 2: Ideation
drafts	Content drafts with checklist	Stage 3: Drafts
captions	Caption variants with A/B tags	Stage 4: Captions
hashtags	Generated hashtags per draft	Stage 5: Hashtags
hashtag_groups	Reusable saved hashtag sets	Stage 5: Hashtags
banned_hashtags	Brand-level banned tags	Stage 5: Hashtags
creative_assets	Images, videos, carousel slides	Stage 6: Creatives

asset_platform_variants	Resized versions per platform	Stage 6: Creatives
brand_templates	Brand overlay templates	Stage 6: Creatives
creative_version_history	Asset version tracking	Stage 6: Creatives
approval_logs	Approval state transitions	Stage 7: Approval
scheduled_posts	Scheduled publishing jobs	Stage 8: Scheduling
best_time_suggestions	Optimal posting times	Stage 8: Scheduling
post_analytics	Performance metrics snapshots	Stage 9: Analytics
post_comments	Comment tracking and replies	Stage 9: Analytics
weekly_reports	Auto-generated weekly reports	Stage 9: Analytics
learning_signals	Performance-based insights	Stage 10: Learning
repurposed_content	Original-to-repurposed links	Stage 10: Learning
notifications	All notification records	Section 6: Notifications
prompt_templates	LLM prompt templates	System-wide
api_usage_logs	API call tracking and costs	System-wide
job_queue	Background job management	System-wide

9. API Endpoints — Master List

Total API endpoints for V1.2: 52+ endpoints across 10 service domains. Detailed request/response schemas are documented in each stage section. All endpoints require JWT authentication. Rate limiting applies per workspace.

Domain	Endpoint Count	Key Operations
Strategy & Pillars	8	Pillar CRUD, competitor management, compliance check
Ideation	6	Generate ideas, manage saved ideas, trending, calendar
Drafts	5	CRUD drafts, checklist management
Captions	7	Generate, adapt, edit, select, A/B tag, preview
Hashtags	6	Generate, manage, groups, banned list
Creative Assets	8	Generate, upload, resize, alt text, templates, versions
Approval	6	Submit, approve, reject, changes, pending list, log
Scheduling	6	Schedule, calendar, reschedule, cancel, best times, conflicts
Analytics	7	Stats, comments, replies, weekly reports, dashboard, A/B results
Learning & Repurposing	3	Signals, repurpose, winners

10. Frontend Component Map

Component	Description	Used In
StrategyHub	Content pillar setup, distribution chart, competitor input	Stage 1
IdeasHub	Dual-panel idea generation + trending topics	Stage 2
PostEditor	Main editor with tabbed sections (copy, hashtags, creative, preview)	Stage 3-6
DraftChecklistWidget	Sticky sidebar showing completion status	Stage 3
CaptionEditor	Rich text editor with char counter + tone selector	Stage 4
PlatformPreviewPanel	Mock-up previews per platform	Stage 4
ABToggle	A/B test variant tagging control	Stage 4
HashtagManager	Tier-organized tag editor with placement options	Stage 5
CreativeGenerator	Image generation interface with template overlay	Stage 6
AssetResizePreview	Shows auto-resized variants per platform	Stage 6
VersionHistoryPanel	Previous generation versions with restore option	Stage 6
ApprovalReviewScreen	Full post preview for approvers with action buttons	Stage 7
BulkCalendar	FullCalendar-based drag-and-drop scheduler	Stage 8
BestTimeSuggestionOverlay	Highlighted time slots on calendar	Stage 8

PostStatsCard	Quick stats badge on published post cards	Stage 9
MessengerInbox	Comment monitoring + reply interface	Stage 9
WeeklyReportView	Interactive weekly analytics report	Stage 9
RepurposePrompt	Modal suggesting format conversion for winners	Stage 10
AdminDashboard	Pipeline status, queue health, API usage, alerts	Admin
NotificationCenter	In-app notification panel with filtering	System

11. Phase Delivery Roadmap

Phase 1 — NOW (Core MVP)

- Onboarding (7 steps as defined in RFC)
- Strategy Hub with content pillars
- Idea generation + trending topics panel
- Draft creation with checklist widget
- Caption generation with platform adaptation + A/B tagging + char count + preview
- Hashtag generation engine (full)
- Image generation + upload + auto-resize + alt text + brand templates
- Video upload (generation in Phase 2)
- Approval pipeline with SLA escalation
- Scheduling with best-time suggestions + bulk calendar + conflict check
- Post-publish quick stats (24h/48h)
- Comment monitoring + AI/human reply
- Weekly report generation
- Content repurposing prompts
- Learning signals extraction
- Messenger module
- Admin dashboard + brand management + permissions
- Full notification system

Phase 2

- Image API integration fully completed (advanced editing)
- Video API integration (AI generation)
- In-platform creative editing for ads
- Competitor insight crawling (advanced)

Phase 3

- Ads generator templates + platform exports
- UGC pack generator
- Email sequences module
- Advanced template marketplace

Phase 4

- Full learning loop automation (auto-suggest next week plan)
- Multi-brand scaling + security hardening
- Slack/WhatsApp notification integrations
- White-label options

12. Acceptance Criteria & QA Checklist

Onboarding Must Pass

Check	Criteria
Workspace created	Workspace exists in DB, user is Owner
Brand created	Brand profile saved with all required fields
Platform connected	At least 1 platform connected + OAuth verified
Brand DNA generated	Brand DNA JSON exists and is editable
Launch plan saved	7-day plan template created

Core Workflow Must Pass

Check	Pass Criteria
Content pillars created	2-10 pillars, percentages sum to 100%
Ideas generated	20 ideas in <8 seconds, correctly tagged to pillars
Trending topics loaded	Panel loads in <3 seconds with relevant topics
Idea saved	Status changes to 'saved', appears in calendar
Draft created from idea	All metadata pre-populated, checklist widget visible
Captions generated	3 variants in <5 seconds, platform-adapted
Character count works	Real-time count with correct color coding per platform
Platform preview renders	Accurate mock-up per platform
A/B tags persist	Tags saved and linked to analytics
Hashtags generated	Correct tier distribution, banned tags excluded
Instagram first-comment option works	Hashtags post as first comment on publish
Image generated	AI image in <15 seconds
Auto-resize works	Correct dimensions for all target platforms
Alt text generated	≤125 chars, contextually relevant
Brand template overlay works	Logo, colors, fonts correctly applied
Version history tracks regenerations	Previous versions accessible
Submit for approval works	Blocked if checklist incomplete (server-side)
Approver notified	Notification within 30 seconds
SLA escalation triggers	12h/24h/48h escalation correct
Approval state machine correct	All transitions logged with timestamps
Scheduling works	Post publishes within 60 seconds of scheduled time

Best time suggestions display	Highlighted slots on calendar
Conflict detection works	Warning for posts <15 minutes apart
Bulk calendar drag-and-drop	Rescheduling without page reload
Publish success	Platform post ID stored, status = published
Publish failure handled	3 retries, then notification sent
24h quick stats display	Stats badge visible on post card
Comments sync	New comments appear within 15 minutes
AI reply works	Reply in <5 seconds, matches Brand DNA tone
Weekly report generates	Auto-generated Sunday, all sections populated
A/B results reported	Correct comparison of tagged variants
Winner detection works	Top 20% flagged within 50 hours
Repurpose prompt appears	Modal with format options for winners
Learning signals extracted	Signals stored within 1h of report generation

— *End of Document* —